

COMIC CRAFT — AI COMIC STORY CREATOR USING GEMINI MODELS

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Date	30 September 2026
Team ID	SWTID-2026-8849
Project Name	Comic Craft-AI
Maximum Marks	4 Marks

Solution Requirements

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority
1	User Input	Collect story prompt, main character name, setting, tone and art style.	High
2	Story Outline	Generate a structured five-panel outline using Gemini Flash.	High
3	Story Generation	Generate narration and dialogue using Gemini Pro.	High
4	Image Generation	Generate comic-style illustrations using Stable Diffusion / Diffusers.	High
5	Preview	Display generated panels, titles, images, scene descriptions and narration.	High
6	PDF Export	Build the layout and save a downloadable multi-page PDF.	High
7	API Access	Provide JSON comic-generation and image-test routes.	Medium
8	Error Handling	Handle API and processing failures with FastAPI error handling.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target / Acceptance Criteria
1	Usability	Simple web form and sequential comic preview.	Clear labels and direct user flow
2	Maintainability	Separate modules for AI, image generation, layout and export.	Modular Python files
3	Reliability	Handle generation failures and invalid processing states.	Try/except and HTTPException
4	Portability	Python-based local setup with requirements.txt.	Python 3.10+ documented
5	Performance	Generation is coordinated as a pipeline.	Performance depends on Gemini and local image generation hardware
6	Security	API keys/tokens are expected through environment configuration.	Do not hard-code secrets